

Database Systems CS-205 (Spring 2023)**Section: BDS-4A, BDS-4B & BDS-4C Due Date: 27/03/2023****Instructions:**

- This assignment is an individual assignment.
- Clearly mention any assumption you have made.
- Please upload your solution in PDF format on google classroom and submit handwritten assignment in classroom with your 8 digit Roll No (19L-1234) as file name.
- Plagiarism would be strictly punished.

Assignment# 3**Total Marks = 100****TOPIC: Functional Dependencies (FDs) (5x6=30 points)**

Q1. Consider the relation R (A, B, C, D, E) and a set of FDs $F = \{C \rightarrow AB, A \rightarrow E, D \rightarrow E, BD \rightarrow C, CD \rightarrow B\}$. Find all possible Keys of R.

Q2. Consider the relation R (A, B, C, D) and a set of FDs $F = \{AB \rightarrow C, CD \rightarrow B, AD \rightarrow B, AC \rightarrow D\}$. Find all possible Keys of R.

Q3. Consider the relation R (A, B, C, D, E) and a set of FDs $F = \{A \rightarrow C, C \rightarrow BD, D \rightarrow A\}$. Find all possible Keys of R.

Q4. Find out whether the following set of functional dependencies for the relation R (A, B, C, D, E, G) are equivalent or not. Show all the steps. $F_1 = \{A \rightarrow C, AB \rightarrow C, C \rightarrow DG, CD \rightarrow G, EC \rightarrow AB, EG \rightarrow C\}$ and $F_2 = \{A \rightarrow C, C \rightarrow D, C \rightarrow G, EC \rightarrow A, EC \rightarrow B, EG \rightarrow C\}$

Q5. Consider the relation R (A, B, C, D, E, G) and a set of FDs $F = \{ABC \rightarrow CDEG, C \rightarrow E, A \rightarrow B, D \rightarrow G\}$. Compute the minimal cover for F (i.e. F'). Also find all possible Keys (i.e. minimal of super keys) of R.

Q6. Consider the relation R (A, B, C, D, E, G) and a set of FDs $F = \{D \rightarrow E, ABC \rightarrow BDE, B \rightarrow G, A \rightarrow C, ABC \rightarrow G\}$. Compute the minimal cover for F (i.e. F'). Also find all possible Keys (i.e. minimal of super keys) of R.

TOPIC: Normalization for Relational Databases

Q1. Consider a relation schema R (A, B, C, D, E, F, G), with FDs $F = \{A \rightarrow G, F \rightarrow E, G \rightarrow DB, D \rightarrow C\}$. Show all steps, working, and reasoning to answer the following questions. **(5x3 = 15 points)**

- Determine all possible keys. Prove it.
- Identify the best normal form that R satisfies (1NF, 2NF, 3NF, or BCNF). Justify your answer.
- Decompose the relation R into a 2NF schema, if it is not in 2NF. (Remove 2NF violations only, in this part)
- Check whether your answer to part (c) is in 3NF. If not, decompose it into a 3NF schema. List clearly complete set of 3NF schema relations with all keys and FDs.
- Check whether your answer to part (d) is in BCNF. If not, decompose it into a BCNF schema. List clearly complete set of BCNF schema relations with all keys and FDs and indicate which dependencies if any are not preserved.

Q2. Consider a relation schema $R(A, B, C, D, E)$, with FDs $F = \{A \rightarrow BC, C \rightarrow D, E \rightarrow D, BE \rightarrow A\}$. Show all steps, working, and reasoning to answer the following questions. **(4x5 = 20 points)**

- Identify the best normal form that R satisfies (1NF, 2NF, 3NF, or BCNF). Justify your answer.
- Decompose the relation R into a 2NF schema if it is not in 2NF. (Remove 2NF violations only, in this part)
- Check whether your answer to part (b) is in 3NF. If not, decompose it into a 3NF schema. List clearly complete set of 3NF schema relations with all keys and FDs.
- Check whether your answer to part (c) is in BCNF. If not, decompose it into a BCNF schema. List clearly complete set of BCNF schema relations with all keys and FDs and indicate which dependencies if any are not preserved.

Q3. Consider the relation $R(A, B, C, D)$, with FDs $F = \{B \rightarrow D, D \rightarrow A\}$. **(5x1=5 points)**

Identify the best normal form that R satisfies (1NF, 2NF, 3NF, or BCNF). Justify your answer. If R is not in BCNF, decompose it until it becomes BCNF. At each step, identify a new relation, decompose and re-compute the keys and the normal forms they satisfy. Indicate which dependencies if any are not preserved by the decomposition.

Q4. Consider the relation $R(A, B, C, D, E)$, with FDs $F = \{A \rightarrow BC, C \rightarrow D, E \rightarrow D, BE \rightarrow A\}$. **(5x3=15 points)**

- Is the decomposition $R_1(A, E)$, $R_2(A, B, C)$, and $R_3(D, E)$ a lossless decomposition and dependency preserving? Prove it.
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- Is the decomposition $R_1(A, B, C)$, $R_2(C, D)$, and $R_3(D, E)$ a lossless decomposition and dependency preserving? Prove it.

Q5. Find out whether the following set of FDs are equivalent or not. Show all the steps. **(5x1 = 5 points)**

$F_1 = \{A \rightarrow C, AB \rightarrow C, C \rightarrow DI, CD \rightarrow I, EC \rightarrow AB, EI \rightarrow C\}$

$F_2 = \{A \rightarrow C, C \rightarrow D, C \rightarrow I, EC \rightarrow A, EC \rightarrow B, EI \rightarrow C\}$

Q6. Consider the relation $R(A, B, C, D, E)$, with FDs $F = \{A \rightarrow BC, C \rightarrow D, E \rightarrow D, BE \rightarrow A\}$. **(5x2 = 10 points)**

- Is the decomposition $R_1(A, E)$, $R_2(A, B, C)$, and $R_3(D, E)$ a lossless decomposition? Prove it.
- Is the decomposition $R_1(A, E)$, $R_2(A, B, C)$, and $R_3(C, D)$ a lossless decomposition? Prove it.